Wearable Technologies and Applications (Wearable Informatics)

Winfree

Lecture 10

Pre-Discussion

At this point, you have:

- Introduced your project
  - With a background review, previous literature / prior work
  - Which has allowed you to frame the importance of your work
- Detailed your hardware design and methods
  - In this case, you have probably included schematics and descriptions of what you have built
  - And, what steps you took in your study
- Provided the results
  - But the results section is typically very dry
  - It *should* be very dry!
  - In the results is *not* where you discuss the interpretation or meaning of your results ... that goes into the discussion section

Purpose

There are five key goals of the discussion section

1. Interpretation
2. Context
3. Limitations
4. Implications
5. Next steps

Structure

Five key goals

1. Interpretation
2. Context
3. Limitations
4. Implications
5. Next steps

Make it flow

- Concise summary of main findings
- Interpretation / Meaning
- Comparison to prior work
- Limitations (those with impact)
- Practical implications
- Future work / Concluding statement

Ask Yourself

But there is more to it than just hitting five bullet points with six parts of the story.

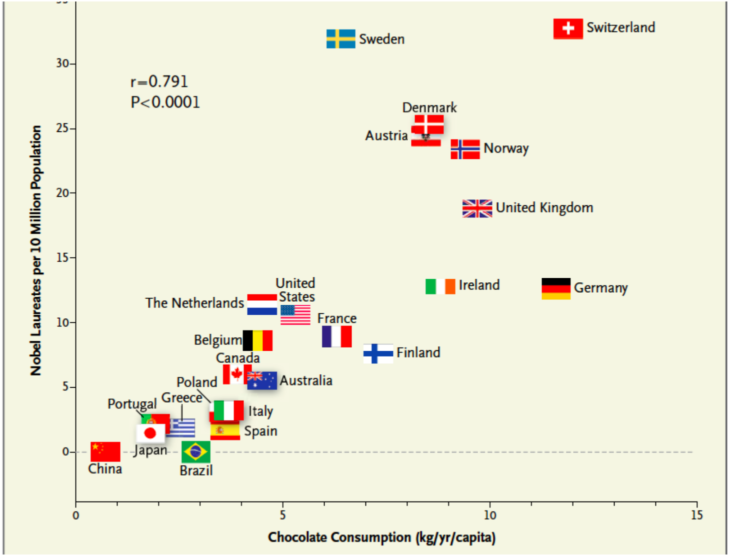
Here are a few tips, or considerations, in writing a clear and convincing discussion section.

Note, much of this will feel like stuff that should have been considered before starting the project or study. Fair point, and yes, much of this is. But not all of this will be obvious ahead of time, or even practical to address at that time.

Results vs Interpretation

Distinguishing results vs interpretation:

- What does the data directly show?
- What are you inferring?
- Avoid overstating causality!



Link Analyses to Claims

Linking analyses to claims:

- Map each major claim to the supporting analysis/figure and quantify uncertainty.
- Make use of confidence intervals, effect sizes, p-values where relevant.

Significance

Effect sizes and practical significance:

- Explain why small statistically significant differences may be meaningless for real-world wearable use.
- Are there clinically significant thresholds that should be considered?

Uncertainty

Uncertainty and robustness checks:

- Consider sensitivity analyses:
  - What happens when you vary assumptions, inputs, model choices, or processing steps?
  - If small, plausible changes produce large shifts in conclusions, those conclusions are fragile.
  - Sensors and prepossessing choices (filter cutoffs, interpolation) can strongly affect derived measures.
  - Model hyper-parameters, feature selection, window sizes, and labeling methods influence performance.
  - Participant sampling, missing data handling, and ground-truth quality can introduce bias.
- Did you perform cross-validation?
- Do error bars in your plots suggest a dependence in your variables is there?
- reproducibility notes.

Limitations

Limitations (technical and experimental):

- What limitations do your sensors have?
- Is there sampling bias? Not just participants, but sampling rate too.
- How does calibration of your sensors or outputs factor in here?
- Is there a possibility of latency issues? Do they matter in your application? Did you address them?
- Do you have an appropriately diverse (age, gender, ethnicity) participant pool?
- **What are your algorithm assumptions?**

Validity

Confounding, bias, and threats to validity:

- What potential is there for measurement error?
- Do you have any selection bias?
- We talked about over fitting earlier - have you tested for this?
- If you are training a model on ground-truth labels or measures, how confident are you with that ground truth?

Failures

How to discuss failures and negative results:

- Be transparent! Yes, there is value to papers that say "this didn't work as expected."
- Try to explain the root causes of these results.
- What did you learn from these results? How do these results contribute to the field of science? Do these results inform future efforts by others?
- What change to your hardware, software, study design, or even analysis should be considered for "next time"?

Prior Work

When relating your work to prior work:

- Be concise.
- Make *fair* comparisons - explain why differences exist (methods, sensors, populations).
- Science advances through thoughtful critique — pointing out limitations, differences, and improvements respectfully and constructively, not through harsh or dismissive attacks.

Visuals

Visual presentation of results: When working on your discussion, reflect on the results and ask:

- Did you choose the right plots?
- Are those plots labeled for clarity?
- Do your scales mislead the reader? Could they?

Writing Style

Writing style and tone: How you write is as important as the content.

- Are you being concise?
- Are you being honest?
- Do you have defensible claims?
- Avoid hedging that obscures meaning or implies stretch conclusions.
- Avoid jargon! And for that matter, acronyms.
  - Acronyms, by and large, are not universal.
  - Consider how those acronyms might have different common meanings for others.
  - Defining acronyms doesn't not absolve you of this issue.
  - Limit not just your use, including how many you define.
  - Always ask yourself if the acronym actually improves your writing, or if it instead distracts the reader from the point you are trying to make.

Ethics

Ethical and user-centered implications: Science is morally neutral, it doesn't care, but you should.

- Does your study present privacy concerns?
- Are there real-world deployment concerns?
- Could your results be easily misinterpreted in an ethically or morally inappropriate way?

Reproducibility

Reproducibility and data or code sharing:

- What can you include as supplementary materials?
- If you are sharing your code, have you practiced good documentation?

Closing Takeaways

Practical engineering takeaways and closing statements:

- How should your results inform product design?
- Or product decisions?
- Or more generally, next experiments?
  (the science is never done)
- If you wanted your readers to remember one thing from your paper, what would it be? Did you get that when you read your discussion? Restate your point as needed...

TLDR;

Honesty is always better than Hype: clear limits and realistic claims increase your paper's (your) credibility.
Always tie your claims to your evidence: point to the exact analysis or figure that supports each statement.
Quantify uncertainty and the practical impact, do not just report p-values.
Be explicit about your assumptions and how they affect conclusions. Model assumptions included.
Failure is informative: explain why something didn't work, that is still valuable science and engineering.
Clarity in visuals (plots, tables, diagrams) and writing wins: A correct, but unclear paper fails to communicate the thoughts. No one will see your genius if you can't communicate it to them.
Consider the audience: Who will be reading your paper; what researchers, clinicians, or designers need to know what you are sharing and how will they likely use the results?

In 10 minutes and in pairs, identify:

1. 3 strengths
2. 3 weaknesses
3. 2 unstated assumptions
4. Bonus: Reword a sentence to strengthen the claim.

After 10 minutes, we will regroup and report back, then move on to the next paper.

Ask yourself

1. Does each main claim cite a specific result or figure?
2. Is uncertainty reported?
3. Are the limitations clearly stated and their impact discussed?
4. Are alternative explanations and confounding variables considered?
5. Is the practical significance discussed in context of real-world use?
6. Are the methods or thresholds that impact interpretation explained, or mentioned?
7. Is the language appropriately cautious? Or have the authors maybe overstated causality?